

Coherence-Dependent Backreaction in Semiclassical and Analog Gravity: Controlled Limits and an Analog-Gravity Test

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We study an effective-field-theory extension of semiclassical gravity in which a coarse-grained coherence functional $\mathcal{C}[g, \rho; \mu]$, built from relative entropy, generalized entropy, and scale-matched local geometric counterterms, generates an informational stress tensor $\Theta_{\mu\nu}$. The construction is presented as a controlled near-equilibrium framework rather than a UV-complete theory. In a 1+1D conformal toy model and in weak-field/Rindler limits, we obtain leading-order expressions that reduce to standard semiclassical behaviour when $\rho \approx \sigma[g]$. We then formulate an acoustic implementation for a Bose-Einstein-condensate sonic horizon. The informational source is fixed by the exact Gaussian-state identity $S(\rho||\sigma[g]) = [\Delta E - T_H \Delta S_{\text{vN}}]/k_B T_H$: relative entropy equals the free-energy excess over the Hawking-temperature reference. Both coherent and energy-matched thermal injections therefore source the informational stress, and at matched injected energy they differ exactly by the von Neumann entropy of the injected occupation. The primary observable is the coherent-minus-thermal differential $\Delta A = A_{\text{coh}} - A_{\text{th}} = \kappa_{\text{eff}} \Delta S_{\text{vN}}$, which is independent of the Hawking temperature at fixed band occupation. For a representative steep-horizon baseline ($T_H \approx 1$ nK, broadband injection at $\hbar\omega \approx 2k_B T_H$, $N = 10^3$ phonons in $M \approx 100$ modes) the model gives $A_{\text{coh}} \approx 1.6 \times 10^{-6}$, $A_{\text{th}} \approx 1.3 \times 10^{-6}$, and $\Delta A \approx 2.3 \times 10^{-7}$. The density-response step is benchmarked to the hydrodynamic compressibility limit, to a static Gross-Pitaevskii/Bogoliubov linear-response kernel (0.6% shift), and to time-dependent 1D Gross-Pitaevskii simulations that validate the kernel to 0.5%; the informational source term itself remains phenomenological, and the same simulations show that ordinary nonlinear response produces a state-dependent background that dominates the differential at baseline amplitude unless discriminated by its spatial and bandwidth signatures. One-at-a-time $\pm 25\%$ scans about a Gross-Pitaevskii-compatible baseline keep ΔA above 5×10^{-8} . A null result below 5×10^{-8} for the differential in this protocol would falsify the acoustic implementation studied here, not the broader EFT proposal by itself. A positive analog result would test the proposed state-geometry feedback structure in an emergent acoustic spacetime, not by itself establish direct gravitational backreaction in astrophysical spacetime.

I. INTRODUCTION

Can analog-gravity experiments test whether geometry responds only to mean stress-energy or also to coarse-grained state mismatch? We study that question within an effective semiclassical framework and use a Bose-Einstein-condensate (BEC) sonic horizon as a concrete conditional test bed.

Our main phenomenological result is an acoustic implementation in which the informational source is fixed by the exact identity $S(\rho||\sigma[g]) = \beta_H \Delta E - \Delta S_{\text{vN}}$ (relative entropy as free-energy excess over the Hawking-temperature reference). Both coherent and energy-matched thermal phonon injections then source the informational stress at the $\delta\rho/\rho_0 \sim 10^{-6}$ level, and they differ exactly by the von Neumann entropy of the injected occupation. The primary falsifiable target of the paper is this coherent-minus-thermal differential, $\Delta A = \kappa_{\text{eff}} \Delta S_{\text{vN}} \approx 2.3 \times 10^{-7}$ at the baseline: a null result below 5×10^{-8} would falsify the specific acoustic implementation studied here.

A. The Physical Question

Semiclassical gravity uses the renormalized expectation value $\langle T_{\mu\nu} \rangle$ as a source in Einstein's equations [1]. That framework is tightly constrained and extremely successful, but it leaves open a narrower EFT question: can coarse-grained state mismatch relative to a geometry-adapted reference state generate additional backreaction terms in controlled near-equilibrium regimes?

Several lines of work motivate that question. Thermodynamic derivations of Einstein's equations [2] and entanglement-based formulations built on relative entropy, modular Hamiltonians, and generalized entropy [3–5] implicate information-theoretic structure in the semiclassical description of spacetime. Penrose [6] and Diósi [7] proposed that gravity may couple to quantum superposition in ways not exhausted by mean stress-energy alone. Relativistic quantum-information studies likewise show that acceleration and curved backgrounds modify coherence properties of quantum states.

B. Our Approach and Main Results

We construct a *coherence functional* $\mathcal{C}[g, \rho; \mu]$ that measures the informational mismatch between a quantum

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state ρ and a geometry-adapted reference state $\sigma[g]$. At the EFT level, variation with respect to the metric defines an *informational stress tensor* $\Theta_{\mu\nu}$ that corrects the semiclassical Einstein equations:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G(\langle T_{\mu\nu} \rangle_\rho + \Theta_{\mu\nu}[g, \rho; \mu]). \quad (1)$$

When $\rho \approx \sigma[g]$, we have $\Theta_{\mu\nu} \approx 0$ and standard semiclassical gravity is recovered. When ρ departs from the chosen reference state, additional state-dependent corrections can appear.

The framework (called *Coherism* in earlier versions of this work; we refer to it below simply as the coherence-functional EFT) yields the following concrete results:

1. **Controlled limits:** In a 1+1D conformal toy model and in weak-field/Rindler limits, we obtain leading-order expressions for $\Theta_{\mu\nu}$ and recover the standard semiclassical limit when $\rho \approx \sigma[g]$.
2. **Geometry-dependent open-system structure:** The main text uses a phenomenological coarse-grained Lindblad ansatz, and Appendix F shows how geometry-dependent GKLS evolution arises in a standard Unruh-DeWitt detector model.
3. **Primary falsification path:** In the acoustic implementation, the exact free-energy form of the relative entropy gives $\delta\rho/\rho_0 \sim 10^{-6}$ for both coherent and energy-matched thermal injection near a sonic horizon; the two differ by the injected von Neumann entropy, giving a differential $\Delta A = \kappa_{\text{eff}} \Delta S_{\text{vN}} \approx 2.3 \times 10^{-7}$ that is independent of the Hawking temperature at fixed band occupation.
4. **Model-dependent null test:** A result below 5×10^{-8} for the coherent-minus-thermal differential would falsify the acoustic implementation studied here.

C. Framework Overview

The basic picture is sketched in Fig. 1. At a given coarse-graining scale μ^{-1} , the metric $g_{\mu\nu}$ induces a family of local reference states $\sigma[g]$ on causal diamonds. The actual quantum state ρ may differ from $\sigma[g]$ in its entanglement and correlation structure. This mismatch is quantified by a local relative-entropy density $s_{\text{rel}}(x; \mu)$ and a generalized entropy density $s_{\text{tot}}(x; \mu)$. Stationarity of $\mathcal{C}$ under variations of $g_{\mu\nu}$ and ρ yields coupled evolution equations: an Einstein-like equation sourced by both $\langle T_{\mu\nu} \rangle$ and an informational stress $\Theta_{\mu\nu}$, and a Lindblad-type equation in which geometry enters through a dissipative term $\mathcal{L}_g[\rho]$ that drives ρ toward compatibility with $\sigma[g]$.

We do not claim a UV-complete theory of quantum gravity, nor a derivation of all known laws from a single principle. This framework is presented as an *effective field theory* (EFT) proposal for a feedback law that may

organize, and slightly deform, the semiclassical frontier between GR and QM. The value of the framework will ultimately be decided by experiment—specifically, by the BEC analog gravity test described in Sec. VI and Appendix E. *Scope:* We present an EFT-level variational framework with one concrete analog-gravity falsification path. The acoustic calculation is intended to test the state-geometry feedback structure of the proposal; it is not, by itself, a demonstration of direct gravitational backreaction in astrophysical spacetime.

The rest of the paper is organized as follows. In Sec. II we review the necessary mathematical tools: causal diamonds, relative entropy, generalized entropy, and semiclassical gravity. In Sec. III we define the 3+1D coherence functional and the informational stress tensor. Sec. IV summarizes a 1+1D toy model that recovers standard limits. Sec. V relates the framework to other approaches including stochastic gravity and entanglement-based gravity. Sec. VI presents the analog-gravity observable, protocol, and reproducibility information. Sec. VII discusses the regime of validity, the main limitations, and the relation to neighboring approaches. Sec. VIII concludes.

II. MATHEMATICAL PRELIMINARIES

In this section we introduce the main mathematical ingredients used in the construction of the coherence functional: causal diamonds, geometry-adapted reference states, relative entropy and generalized entropy densities, and the semiclassical Einstein equation.

A. Causal diamonds and local algebras

We work on a globally hyperbolic spacetime $(\mathcal{M}, g_{\mu\nu})$ with signature $(+, -, -, -)$ and set $c = \hbar = k_B = 1$. Given a point $x \in \mathcal{M}$ and a length scale $\ell \sim \mu^{-1}$, we consider a *causal diamond* $\mathcal{D}(x; \mu)$ defined as the intersection of the chronological future of a point p^- and the chronological past of a point p^+ such that x lies on a timelike geodesic between them and the proper time separation is 2ℓ . On each diamond we restrict the field algebra to a von Neumann algebra $\mathcal{A}(\mathcal{D})$.

For a global quantum state ρ on the full field algebra, we denote by $\rho_{\mathcal{D}}$ the reduced state obtained by tracing out degrees of freedom outside $\mathcal{D}(x; \mu)$. We similarly define a reference state $\sigma[g]$ that depends functionally on the metric $g_{\mu\nu}$ and restrict it to $\mathcal{D}$ as $\sigma_{\mathcal{D}}[g]$.

a. What fixes the reference state $\sigma[g]$ (operationally). In the controlled examples in this paper, $\sigma[g]$ is chosen as the standard geometry-adapted Hadamard/KMS state for the relevant causal diamond: Minkowski vacuum in flat space, Unruh/Hartle–Hawking-type vacua near horizons, and a thermal KMS state at the acoustic Hawking temperature for the BEC sonic horizon. The coarse-graining scale μ sets the diamond size and the smearing used to define $s_{\text{rel}}(x; \mu)$; scheme dependence in absolute values

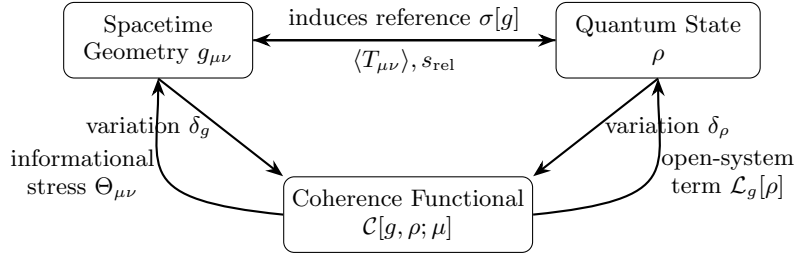


FIG. 1. Schematic feedback loop. At a coarse-graining scale μ , the geometry $g_{\mu\nu}$ defines a family of reference states $\sigma[g]$ on local causal diamonds. The mismatch between ρ and $\sigma[g]$ enters the coherence functional $\mathcal{C}[g, \rho; \mu]$, whose variation yields an informational stress tensor $\Theta_{\mu\nu}$ and a geometry-dependent open-system generator $\mathcal{L}_g[\rho]$.

is treated as a coarse-graining ambiguity, while physical predictions are organized around controlled limits (and, in particular, around variations and near-equilibrium expansions).

B. Relative entropy and generalized entropy

Given two states ρ and σ on the same algebra, the quantum relative entropy is

$$S(\rho||\sigma) = \text{tr}(\rho \log \rho) - \text{tr}(\rho \log \sigma) \geq 0. \quad (2)$$

Relative entropy is a measure of distinguishability. It is monotonic under partial trace (data processing inequality) and invariant under unitary evolution.

On a causal diamond $\mathcal{D}(x; \mu)$ we define a *smear relative-entropy density* via a coarse-graining procedure. Let $\{\mathcal{D}_i\}$ be a covering of spacetime by diamonds of characteristic size μ^{-1} , and let $f_i(x)$ be a partition of unity subordinate to this covering. We define

$$s_{\text{rel}}(x; \mu) \equiv \sum_i f_i(x) \cdot \frac{1}{V_{\mathcal{D}_i}} S(\rho_{\mathcal{D}_i} || \sigma_{\mathcal{D}_i}[g]), \quad (3)$$

where $V_{\mathcal{D}_i}$ is the four-volume of each diamond.

Important caveat: Relative entropy is fundamentally a non-local quantity defined on subsystems, not a true local density. The object $s_{\text{rel}}(x; \mu)$ should be understood as a coarse-grained, scale-dependent quantity whose integral over spacetime is well-defined, but whose point-wise values depend on the choice of covering and partition of unity. Physical predictions must be independent of these choices in the continuum limit $\mu \rightarrow \infty$, which we verify in explicit examples below. This construction is analogous to the smearing used in algebraic QFT to define local observables from field operators [8].

We also use a coarse-grained *generalized entropy density* inspired by the black-hole [9] and holographic literature [5]. Schematically,

$$s_{\text{tot}}(x; \mu) \equiv \frac{1}{V_{\mathcal{D}}} (S_{\text{vN}}[\rho_{\mathcal{D}}] + \frac{\text{Area}[\partial\mathcal{D}]}{4G_{\text{eff}}(\mu)}), \quad (4)$$

where S_{vN} is the von Neumann entropy of the reduced state, $\text{Area}[\partial\mathcal{D}]$ is the area of the boundary of the dia-

mond, and $G_{\text{eff}}(\mu)$ encodes the scale dependence of the gravitational coupling and higher-curvature terms.

Both s_{rel} and s_{tot} depend on the choice of reference state $\sigma[g]$ and the coarse-graining scale μ , but are otherwise standard objects in quantum field theory on curved spacetimes.

C. Semiclassical Einstein equation

In semiclassical gravity the metric is taken as classical, while matter is treated quantum mechanically. The central equation is

$$G_{\mu\nu}[g] + \Lambda g_{\mu\nu} = 8\pi G \langle T_{\mu\nu} \rangle_\rho, \quad (5)$$

where $G_{\mu\nu}$ is the Einstein tensor, Λ the cosmological constant, G Newton's constant, and $\langle T_{\mu\nu} \rangle_\rho$ the renormalized expectation value of the stress-energy tensor in the state ρ .

Null-limit checklist (consistency limits).

- $\rho = \sigma[g] \Rightarrow \Theta_{\mu\nu} = 0$ (equilibrium / no informational mismatch).
- Flat space with no coarse-graining ($\mu \rightarrow \infty$) $\Rightarrow$ geometric/coherence corrections vanish.
- Weak field / near-equilibrium $\Rightarrow$ semiclassical Einstein equation (and Einstein–Langevin structure) recovered.
- 1+1D conformal case $\Rightarrow$ Polyakov correspondence and stochastic-gravity limits recovered.

This equation can be generalized to include stress-energy fluctuations in the Einstein–Langevin form [1], where a stochastic source term encodes quantum fluctuations. We will not need the full machinery here, but we will require that our informational stress tensor $\Theta_{\mu\nu}$ be consistent with the conservation law

$$\nabla^\mu (\langle T_{\mu\nu} \rangle_\rho + \Theta_{\mu\nu}) = 0, \quad (6)$$

so that the contracted Bianchi identity remains satisfied.

D. Open quantum systems and Lindblad generators

The evolution of ρ in the presence of an environment, or under coarse-graining, is often described by a Lindblad (GKLS) master equation [10]

$$\dot{\rho} = -i[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right), \quad (7)$$

where $\{L_k\}$ are Lindblad operators. This evolution is trace-preserving, completely positive, and Markovian. In our setting, the geometry $g_{\mu\nu}$ will enter both through the Hamiltonian $H[g]$ and, in an effective way, through the dissipative term that tends to reduce the mismatch between ρ and $\sigma[g]$.

III. THE COHERENCE FUNCTIONAL IN 3+1 DIMENSIONS

We now define the coherence functional $\mathcal{C}[g, \rho; \mu]$ in 3+1 dimensions using the ingredients introduced above. Our guiding principle is that $\mathcal{C}$ should (i) be diffeomorphism invariant, (ii) reduce in appropriate limits to familiar gravitational and entropic quantities, and (iii) yield equations of motion that are compatible with conservation and known semiclassical behaviour.

A. Definition of the functional

We consider a family of causal diamonds $\mathcal{D}(x; \mu)$ that cover the spacetime at a coarse-graining scale μ . On each diamond, we evaluate the relative-entropy density $s_{\text{rel}}(x; \mu)$ and generalized entropy density $s_{\text{tot}}(x; \mu)$ defined in Eqs. (3) and (4). Because s_{rel} and s_{tot} are coarse-grained four-volume densities, the geometric term entering the same integrand must also scale as L^{-4} . We therefore include a local geometric density $\mathcal{R}(x; \mu)$ in which the Einstein–Hilbert term is accompanied by an explicit inverse-length scale:

$$\mathcal{R}(x; \mu) = M_g^2(\mu) R(x) + \lambda_1(\mu) R^2 + \lambda_2(\mu) R_{\alpha\beta} R^{\alpha\beta} + \dots, \quad (8)$$

where R is the Ricci scalar, $M_g^2(\mu)$ is a running coefficient with dimensions of inverse length squared, and the $\lambda_i(\mu)$ are dimensionless higher-curvature Wilson coefficients. This makes $\mathcal{R}$, s_{rel} , and s_{tot} all scale as L^{-4} inside the four-dimensional integrand.

The coherence functional is then defined as

$$\mathcal{C}[g, \rho; \mu] = \int_{\mathcal{M}} d^4x \sqrt{-g} \left[\alpha s_{\text{rel}}(x; \mu) - \beta s_{\text{tot}}(x; \mu) + \gamma \mathcal{R}(x; \mu) \right], \quad (9)$$

where α, β, γ are dimensionless coefficients. With Eq. (8), every term in the integrand has the same density dimension, so the functional is well-posed as an EFT variational object. The signs are chosen so that:

- the term proportional to α rewards alignment between ρ and $\sigma[g]$ (low relative entropy),
- the term proportional to β implements an entropy constraint or cost, and
- the term proportional to γ reproduces the Einstein–Hilbert contribution once the running scale $M_g^2(\mu)$ is restored.

The choice of $\sigma[g]$ is not unique, and we now specify our construction explicitly.

Regime of validity: We restrict to the regime where causal diamonds satisfy $L \ll L_{\text{curvature}}$, i.e., diamond size much smaller than the local radius of curvature. In this limit, the geometry inside each diamond is approximately flat (Riemann normal coordinates), and we can leverage results from flat-space algebraic QFT.

Explicit construction: For a diamond $\mathcal{D}$ in this regime, we define $\sigma_{\mathcal{D}}[g]$ as the *vacuum state restricted to the diamond*, where “vacuum” means the Hadamard state [11] that reduces to the Minkowski vacuum in the local inertial frame. Concretely:

1. Construct Riemann normal coordinates centered on the diamond.
2. Define $\sigma_{\mathcal{D}}[g]$ as the restriction to $\mathcal{D}$ of the state whose two-point function matches the Hadamard form to order $O(L^2/L_{\text{curvature}}^2)$.
3. The modular Hamiltonian $K_{\sigma} = -\log \sigma_{\mathcal{D}}$ then approaches the boost generator for the diamond [4], with corrections of order $(L/L_{\text{curvature}})^2$.

This construction is well-defined for conformally coupled scalar fields and can be extended to other fields satisfying the microlocal spectrum condition.

Limitations: For diamonds comparable to or larger than the curvature scale, or in regions with strong time-dependence (e.g., near cosmological horizons), this construction requires modification. We do not address these regimes in this paper.

Dependence on choice: Different choices of $\sigma[g]$ (e.g., different Hadamard states) will yield different values of s_{rel} . However, the *difference* in relative entropy between two physical states ρ_1 and ρ_2 relative to the same $\sigma[g]$ is independent of this choice to leading order, since the UV-divergent parts cancel. Physical predictions involving $\Delta s_{\text{rel}} = s_{\text{rel}}[\rho_1] - s_{\text{rel}}[\rho_2]$ are therefore robust. Predictions involving absolute values of s_{rel} should be interpreted with this ambiguity in mind.

B. Informational Stress Tensor

The variation of $\mathcal{C}$ with respect to the metric defines the informational stress tensor $\Theta_{\mu\nu}$:

$$\Theta_{\mu\nu}[g, \rho; \mu] \equiv -\frac{2}{\sqrt{-g}} \frac{\delta \mathcal{C}}{\delta g^{\mu\nu}}. \quad (10)$$

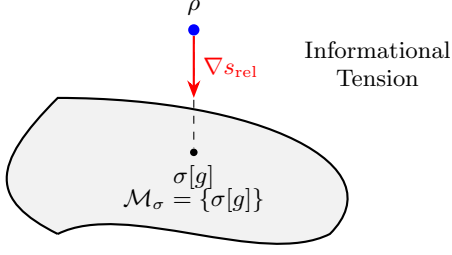


FIG. 2. Geometric interpretation of informational tension. The manifold $\mathcal{M}_\sigma$ represents the set of geometry-adapted reference states. The actual state ρ is displaced from this manifold. The gradient of the relative entropy s_{rel} acts as a restoring force (informational tension) that drives both the geometry (modifying $\sigma[g]$) and the state (via $\mathcal{L}_g$) to reduce the mismatch.

Expanding the functional $\mathcal{C} = \alpha S_{\text{rel}} - \beta S_{\text{gen}} + \dots$ yields a leading term proportional to the modular-Hamiltonian variation:

$$\Theta_{\mu\nu} \approx \kappa(\mu) \frac{1}{V_D} \left\langle \frac{\delta K_\sigma}{\delta g^{\mu\nu}} \right\rangle_\rho. \quad (11)$$

Here $\kappa(\mu)$ is an implementation-dependent EFT coefficient that absorbs convention-dependent $O(1)$ normalizations from the coarse-graining prescription. In the present paper we do not claim a unique universal first-principles value for κ in generic 3+1D settings. Crucially, if the functional $\mathcal{C}$ is constructed to be invariant under spacetime diffeomorphisms (a requirement for any covariant theory), Noether's second theorem guarantees that the total stress tensor is covariantly conserved:

$$\nabla^\mu (\langle T_{\mu\nu}^{\text{mat}} \rangle_\rho + \Theta_{\mu\nu}) = 0. \quad (12)$$

This conservation law ensures the consistency of the augmented Einstein equations. Formally, this yields three contributions,

$$\Theta_{\mu\nu} = \alpha \Theta_{\mu\nu}^{\text{rel}} - \beta \Theta_{\mu\nu}^{\text{ent}} + \gamma \Theta_{\mu\nu}^{\text{geom}}, \quad (13)$$

coming from the relative-entropy, generalized entropy, and geometric parts of the integrand, respectively.

The explicit form of the first two terms is nontrivial, since s_{rel} and s_{tot} depend on $g_{\mu\nu}$ both through the induced algebras and through the reference state $\sigma[g]$.

What we derive vs. what we assume: In this paper, we obtain controlled leading-order expressions for $\Theta_{\mu\nu}$ in the following settings:

1. 1+1D conformal case (Sec. IV, Appendix C)
2. 3+1D weak-field/Rindler limit (Appendix B, D)
3. acoustic implementation for a BEC sonic horizon (Appendix E)

For general curved spacetimes outside these regimes, the detailed form of $\Theta_{\mu\nu}$ is an EFT assumption constrained by covariance, conservation, and the semiclassical limit. In the acoustic sector specifically, the part benchmarked to standard BEC dynamics is the linear density response once a weak source $\Theta_{tt}^{(\text{analog})}$ is specified; the source amplitude κ_{eff} , the detailed source form, and the matched thermal-control identification remain implementation assumptions spelled out in Appendix E.

Explicit weak-field result: In the regime $|h_{\mu\nu}| \ll 1$ with $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, the leading contribution to the informational stress tensor from a coherent excitation is (see Appendix D for derivation):

$$\Theta_{\mu\nu} = \frac{\alpha}{V_D} (\langle T_{\mu\nu} \rangle_\rho - \langle T_{\mu\nu} \rangle_\sigma) + O(h^2, L_P^2/L^2), \quad (14)$$

where the first term is the difference in stress-energy expectation values between the actual and reference states, and the corrections are suppressed by metric perturbations and the ratio of Planck to diamond scales.

We impose the following consistency requirements:

1. **Covariance:** $\Theta_{\mu\nu}$ transforms as a rank-2 tensor under diffeomorphisms.
2. **Conservation:** on solutions of the coupled equations,

$$\nabla^\mu (\langle T_{\mu\nu} \rangle_\rho + \Theta_{\mu\nu}) = 0. \quad (15)$$

3. **UV consistency:** divergences in s_{rel} and s_{tot} are absorbed into the local curvature counterterms in $\mathcal{R}$, so that $\Theta_{\mu\nu}$ is finite and renormalized at scale μ .
4. **Semiclassical limit:** in regimes where ρ is locally indistinguishable from $\sigma[g]$ and the generalized entropy is extremal, $\Theta_{\mu\nu}$ reduces to familiar anomaly-induced and semiclassical corrections, and Eq. (5) is recovered.

The entropic field equation is then

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G (\langle T_{\mu\nu} \rangle_\rho + \Theta_{\mu\nu}[g, \rho; \mu]). \quad (16)$$

When $\Theta_{\mu\nu}$ is negligible, this reduces to the standard semiclassical Einstein equation. When $\Theta_{\mu\nu}$ is small but nonzero, one may look for subtle, scale-dependent deviations from GR.

C. Geometry-dependent open-system evolution

Variation of $\mathcal{C}$ with respect to ρ yields a dual equation of motion for the state. Restricting ourselves to Markovian dynamics for simplicity, we write

$$\frac{d\rho}{dt} = -i[H[g], \rho] + \mathcal{L}_g[\rho], \quad (17)$$

where the Lindblad-like term $\mathcal{L}_g[\rho]$ is chosen so that $\mathcal{C}[g; \rho; \mu]$ is stationary or decreases along solutions, subject to suitable constraints. At a heuristic level, $\mathcal{L}_g$ drives ρ toward the geometry-adapted reference state $\sigma[g]$ while preserving trace and complete positivity. The specific form of $\mathcal{L}_g$ is determined by the requirement that ρ relaxes toward $\sigma[g]$ to minimize the informational tension. A natural *phenomenological ansatz* is a double-commutator form driven by the modular Hamiltonian difference $\Delta K = K_\rho - K_\sigma$:

$$\mathcal{L}_g[\rho] = -\Gamma[\Delta K, [\Delta K, \rho]], \quad (18)$$

where $\Gamma > 0$ is a relaxation rate.

Status of this ansatz: We emphasize that Eq. (18) is a phenomenological proposal, not a derivation from first principles. While double-commutator forms generically preserve trace ($\text{tr}(\mathcal{L}_g[\rho]) = 0$), complete positivity (CP) is not automatically guaranteed for arbitrary ΔK . Standard Lindblad theory requires $\mathcal{L}$ to be of GKLS form with positive Lindblad operators; for Eq. (18) to satisfy CP, we require ΔK to be Hermitian (satisfied by definition) and $\Gamma > 0$. A rigorous proof of complete positivity for infinite-dimensional field-theoretic systems is beyond our scope; we assume CP holds in the coarse-grained, finite-dimensional truncation relevant to macroscopic experiments.

Causality concerns: The Reeh-Schlieder theorem implies that local operations in QFT cannot perfectly localize states to bounded regions, potentially conflicting with strict causality for any local dissipative dynamics [8]. We address this as follows:

1. The Lindblad term $\mathcal{L}_g$ acts at the coarse-graining scale μ^{-1} , not at arbitrarily short distances. Violations of strict locality are exponentially suppressed on scales larger than μ^{-1} .
2. For macroscopic experiments ($L \gg L_P$), the effective decoherence rate $\Gamma \sim (L_P/L)^n$ (with $n \geq 2$ on dimensional grounds) ensures that causality violations are unobservably small.
3. A fully UV-complete theory would likely require a non-Markovian, history-dependent generator. The Markovian approximation is valid only for timescales $\tau \gg \mu^{-1}$.

For the macroscopic applications in this paper, these caveats do not affect predictions, but they must be addressed in any UV completion.

A schematic representation of the geometry-state coupling is shown in Fig. 3. Causal diamonds act as local “patches” where geometry, state, and their mismatch are evaluated and fed back into the dynamics.

IV. TOY MODELS AND STANDARD LIMIT RECOVERIES

Before turning to new predictions, we must verify that the entropic framework reproduces known limits. In this

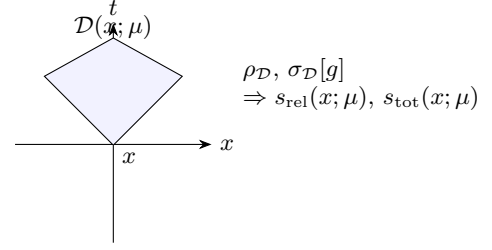


FIG. 3. A causal diamond $\mathcal{D}(x; \mu)$ centered on a point x , with temporal size set by the coarse-graining scale μ^{-1} . The reduced state $\rho_{\mathcal{D}}$ and reference state $\sigma_{\mathcal{D}}[g]$ on the diamond define local relative-entropy and generalized-entropy densities that enter the coherence functional.

section we briefly summarize a 1+1-dimensional toy model in which the informational stress tensor can be written explicitly and shown to reduce to the Polyakov anomaly and Einstein–Langevin form in appropriate regimes. Details are given in Appendix C.

A. 1+1D setup and Polyakov action

In two dimensions, the effective action of a conformal field with central charge c on a background metric $g_{\mu\nu}$ can be written as the Polyakov action [12]

$$S_P[g] = -\frac{c}{96\pi} \int d^2x \sqrt{-g} R \frac{1}{\square} R, \quad (19)$$

whose variation yields the well-known anomaly-induced stress-energy tensor. In this context, one can choose a reference state $\sigma[g]$ whose modular Hamiltonian has a simple local form, and evaluate the relative-entropy and generalized-entropy densities explicitly.

A specific choice of coherence functional in 1+1D can be written as

$$\mathcal{C}_{1+1}[g, \rho] = \alpha S(\rho \| \sigma[g]) - \beta S_{\text{vN}}[\rho] + \gamma S_P[g], \quad (20)$$

where we suppress the explicit coarse-graining scale for simplicity.

Variation with respect to $g_{\mu\nu}$ produces an informational stress tensor of the form

$$\Theta_{\mu\nu} = \kappa_{1+1} [(\beta - \alpha) \Delta \langle T_{\mu\nu} \rangle - \alpha T_{\mu\nu}^{\text{anom}}], \quad (21)$$

where $\Delta \langle T_{\mu\nu} \rangle$ is the difference between the stress-energy in ρ and $\sigma[g]$, $T_{\mu\nu}^{\text{anom}}$ is the anomaly-induced stress, and κ_{1+1} packages the normalization convention of the reduced 1+1D model. The coefficients α, β can then be tuned so that the usual semiclassical contribution and anomaly term are recovered in the limit where ρ tracks $\sigma[g]$.

B. Einstein–Langevin limit

The Einstein–Langevin equation in 1+1D couples the metric to both the expectation value and stochastic fluctuations of the stress-energy tensor. In the entropic toy

model, fluctuations in $\Delta\langle T_{\mu\nu} \rangle$ and the relative entropy can be shown to induce stochastic variations in $\Theta_{\mu\nu}$ with *similar structure* to the Einstein-Langevin noise, at least in Gaussian approximations.

Precise status of this correspondence: We do not claim exact equivalence between our framework and stochastic gravity. Rather, we show that: (i) the *mean* of $\Theta_{\mu\nu}$ matches the semiclassical stress in the limit $\rho \rightarrow \sigma[g]$; and (ii) the *variance* of $\Theta_{\mu\nu}$ has the same tensorial structure as the noise kernel $N_{\mu\nu\alpha\beta}$. A rigorous proof of equivalence would require showing that the two-point function $\langle \Theta_{\mu\nu}(x)\Theta_{\alpha\beta}(y) \rangle$ equals $N_{\mu\nu\alpha\beta}(x, y)$ exactly, which we have not done.

Subject to these caveats, the 1+1D toy model is *consistent* with the standard semiclassical and stochastic limits in appropriate regimes. This supports the interpretation of $\Theta_{\mu\nu}$ as an informational correction to the semiclassical stress, rather than a replacement of it.

C. Relation to Einstein-Langevin Noise

The stochastic gravity framework [1] describes metric fluctuations via the Einstein-Langevin equation. In the present framework, these fluctuations arise deterministically from the informational stress tensor. The noise kernel $N_{abcd}(x, y)$ in stochastic gravity is related to the anticommutator of stress tensor fluctuations. In our framework, we identify this with the variance of $\Theta_{\mu\nu}$:

$$N_{abcd}(x, y) \sim \langle \{ \Theta_{\mu\nu}(x)_{ab}, \Theta_{\mu\nu}(y)_{cd} \} \rangle_{\rho}. \quad (22)$$

This suggests that the “stochastic” noise is actually the signature of the underlying coherent dynamics of the geometry-state coupling.

D. Coherence conservation inequality

Under an additional dissipativity assumption on the coarse-grained Lindblad generator $\mathcal{L}_g[\rho]$, the coherence functional is expected to obey a monotonicity inequality along solutions,

$$\frac{d}{dt} \mathcal{C}[g(t), \rho(t); \mu] \leq 0, \quad (23)$$

with equality only when ρ is locally indistinguishable from $\sigma[g]$ and the generalized entropy is extremal. In the present manuscript we treat Eq. (23) as a plausible coarse-grained consistency condition rather than a fully derived theorem for the complete field-theoretic dynamics. Under that assumption, $\mathcal{C}$ behaves as a Lyapunov-like functional for the geometry–state feedback dynamics.

V. RELATIONS TO OTHER APPROACHES

Here we briefly summarize how this framework relates to several existing lines of research.

a. Priority and relation (one-paragraph summary).

The present manuscript should be read as a phenomenological extension within the existing gravity-from-information literature, not as a clean break from it. The borrowed machinery is the relative-entropy / modular-Hamiltonian / generalized-entropy framework familiar from the causal-diamond and holographic literature [3–5]. The new ansatz introduced here is to package those quantities, together with a scale-matched local geometric density, into an EFT coherence functional whose variation defines an additional mean stress term $\Theta_{\mu\nu}$. Relative to stochastic gravity [1], the paper studies a state-dependent *mean* correction rather than a noise kernel alone. Relative to Penrose–Diósi collapse models [6, 7], it does not introduce an objective collapse law. The narrow experimental novelty is the coherent-versus-thermal control protocol in the analog section, not the existence of sonic horizons or GKLS dynamics by themselves.

Stochastic gravity: the Einstein–Langevin equation already extends semiclassical gravity by incorporating stress–energy fluctuations. This framework is compatible with this and can be viewed as adding an informational contribution to the mean stress.

Entanglement-based gravity: derivations of linearized Einstein equations from entanglement entropy and the “first law” of entanglement motivate the use of relative entropy and generalized entropy. This framework fits naturally into that landscape by promoting those quantities into a variational principle. The connection is motivational rather than derived here: in controlled near-equilibrium settings, relative entropy and modular Hamiltonians provide the natural language for organizing the informational correction.

Penrose–Diósi collapse: these models connect gravitational self-energy of superpositions to objective collapse of the wavefunction. This framework does not introduce a collapse mechanism, but it shares the intuition that coherence interacts nontrivially with gravity.

Relativistic quantum information: [13] studies of non-inertial observers, entanglement in curved spacetimes, and quantum fields in an expanding universe provide concrete setups where geometry modifies quantum coherence. These are natural testbeds for the entropic informational stress tensor.

VI. ANALOG-GRAVITY TEST

The concrete falsification path studied in this paper is an analog-gravity experiment in a Bose-Einstein condensate sonic horizon. Long-baseline gravitational extrapolations are not used in the main argument.

a. Primary observable. For each injected state, Appendix E defines

$$A \equiv \max_{|r-r_H| \leq L_{\text{coh}}} \left| \frac{\delta\rho}{\rho_0} \right|, \quad (24)$$

the maximum fractional density modulation within one coherence length of the sonic horizon. Because the informational source is the exact free-energy form of the relative entropy, $S(\rho||\sigma[g]) = \beta_H \Delta E - \Delta S_{\text{vN}}$, both the coherent injection and the energy-matched thermal control source the informational stress: at the baseline, $A_{\text{coh}} \approx 1.6 \times 10^{-6}$ and $A_{\text{th}} \approx 1.3 \times 10^{-6}$. The primary observable is the differential

$$\Delta A \equiv A_{\text{coh}} - A_{\text{th}} = \kappa_{\text{eff}} \Delta S_{\text{vN}} g_{\text{peak}} \approx 2.3 \times 10^{-7}, \quad (25)$$

which isolates the entropy-sensitive part of the source and is independent of the Hawking temperature at fixed band occupation. A differential below 5×10^{-8} for the parameter regime of Fig. 5 would falsify the acoustic implementation studied in this paper.

b. Null-test assumptions. That threshold is conditional on the assumptions summarized in Appendix E: the injected band remains in the hydrodynamic regime $k\xi \ll 1$, the background condensate response is linear in the weak informational source, the coherent and thermal injections are matched in energy, band, and background geometry, and the injected-band occupation (hence ΔS_{vN}) is independently characterized. The density-response step is benchmarked to the long-wavelength compressibility limit, to a static GP/BdG kernel, and to time-dependent 1D Gross-Pitaevskii simulations (Appendix E); the informational source itself remains an implementation assumption. The same time-dependent simulations show that ordinary Gross-Pitaevskii nonlinearity produces a state-dependent background differential that exceeds ΔA at baseline injection amplitude; discriminating the informational component requires its static, horizon-pinned spatial profile and its ΔS_{vN} bandwidth scaling, and this is the dominant open systematic of the protocol. Under those conditions, a null result would falsify only the acoustic implementation studied here.

c. Interpretational scope. A positive result in the analog system would test the state-geometry feedback structure of the EFT in an emergent acoustic spacetime. It would not, by itself, establish direct gravitational backreaction in astrophysical spacetime.

d. Reproducibility (repro pack). The manuscript and all figures generated by the included scripts are archived at [10.5281/zenodo.17868263](https://zenodo.org/record/17868263) [14]. Figure 5 is reproduced by running `python3 bec_sonic_horizon_simulation.py`, and Fig. 4 by running `python3 gpe_protocol_simulation.py`, in the `physics/` directory (Python 3, NumPy, Matplotlib; < 1 minute each on a laptop).

VII. DISCUSSION

The present manuscript should be read as an EFT proposal with one controlled phenomenological target rather than as a general 3+1D theory of quantum gravity.

a. What is established here. The paper defines a scale-matched coarse-grained functional, exhibits controlled limits in which the informational stress can be written explicitly, and formulates one concrete analog-gravity protocol with a falsification threshold. The weak-field/Rindler and 1+1D constructions show that the framework can be matched to familiar semiclassical structures in near-equilibrium settings. Within the acoustic implementation, the coherent-versus-thermal structure of the prediction is now *derived* from the exact free-energy form of the relative entropy, Eq. (E8), rather than assumed: both injections source the informational stress, and the differential measures the injected von Neumann entropy. The density-response step is validated against time-dependent 1D Gross-Pitaevskii dynamics at the sub-percent level.

b. What is not established here. Several important ingredients remain open. We do not provide a full renormalized existence proof for $\Theta_{\mu\nu}$ in arbitrary 3+1D curved spacetimes. The overall coupling entering phenomenology is treated as implementation dependent rather than universally derived. The numerical model in Appendix E benchmarks the response step statically and dynamically, but it does not derive the informational source term microscopically, and the time-dependent study shows that ordinary Gross-Pitaevskii nonlinearity produces a state-dependent background that dominates the informational differential at baseline injection amplitude; the discrimination strategy (static horizon-pinned profile, ΔS_{vN} bandwidth scaling) is identified but not yet demonstrated in a full simulation of the vortex geometry. For the same reasons, a positive analog result would support the feedback structure of the EFT, but it would not by itself prove direct gravitational backreaction in nature.

c. Relation to neighboring approaches. Relative to stochastic gravity [1], the present proposal introduces a state-dependent mean correction rather than a noise kernel alone. Relative to Penrose–Diósi collapse models [6, 7], it does not invoke objective collapse. Relative to thermodynamic-gravity and entanglement-equilibrium programs [2–5, 15], it packages the same entropic ingredients into a coarse-grained EFT ansatz rather than deriving Einstein dynamics from a thermodynamic extremum principle; the off-equilibrium direction explored here is closest in spirit to the non-equilibrium extension of spacetime thermodynamics proposed by Eling, Guedens, and Jacobson [16]. These distinctions are clearest in the analog protocol, where the control parameter is the free-energy mismatch between the injected state and the geometry-fixed reference state.

d. Consistency constraints from hybrid classical–quantum dynamics. Structurally, the framework pairs a classical metric sourced by a state functional (the $\Theta_{\mu\nu}$ sector) with a geometry-dependent dissipator (the $\mathcal{L}_g$ sector). This places it in the family of consistent hybrid classical–quantum dynamics studied by Kafri, Taylor, and Milburn [17], by Tilloy and Diósi [18], and in the postquantum classical-gravity program of Oppenheim and

collaborators [19, 20]. A central lesson of that literature is that the sourcing and the decoherence cannot be chosen independently: complete positivity and the absence of superluminal signaling [21] enforce a trade-off in which the product of the sourcing strength and the decoherence rate is bounded below. In the present notation, this implies that the coefficient α entering $\Theta_{\mu\nu}$ and the relaxation rate Γ in Eq. (18) must satisfy a consistency relation rather than being independent dials. Deriving the precise α – Γ locking condition for the modular-Hamiltonian dissipator used here—the analog, in our setting, of the decoherence-versus-diffusion trade-off of Ref. [20]—is, in our view, the single most important open theoretical task for the framework: it would reduce the parameter freedom, subject the proposal to existing heating and decoherence bounds in a controlled way, and convert the two ansätze of this paper into one constrained structure.

e. Equivalence-principle structure. If geometry responds to any state functional beyond $\langle T_{\mu\nu} \rangle$, then two systems with equal stress-energy but different coherence gravitate differently, and some state-dependence of free fall is a structural consequence rather than an optional feature. We state this openly and treat it as a constraint channel. The weak-field benchmark of Appendix D gives a coherence-dependent acceleration fraction $\eta_{\text{coh}} \sim 10^{-30} \alpha$ for laboratory-scale coherent states, far below current bounds from atom-interferometric tests of the equivalence principle with superposed and entangled states, $\eta \lesssim 10^{-12}$ [22], and with internal-state superpositions, $\eta \lesssim 5 \times 10^{-8}$ [23]. Such tests therefore do not yet constrain $\alpha \sim 1$, but they define the long-term gravitational falsification channel of the framework; the analog protocol of Sec. VI is the near-term arena precisely because the acoustic coupling κ_{eff} is not suppressed by $(L_P/L)^2$.

f. Hadamard states as attractors (conjecture). If the monotonicity property Eq. (23) holds for the coupled dynamics, the geometry-adapted Hadamard/KMS states $\sigma[g]$ are not merely a kinematical choice but dynamical attractors of the geometry–state feedback: physical states relax toward the Hadamard class. This would promote the Hadamard condition from an axiom of QFT in curved spacetime to a selection effect of the dynamics. We flag this as a conjecture worth separate investigation, since it is the clearest conceptual payoff the framework would deliver if the Lyapunov property can be established.

g. Sign of the informational stress. The sign of $\Theta_{\mu\nu}$ is not fixed universally by the present EFT. In the controlled Rindler example the leading energy-density contribution is positive, which is consistent with the intuition that maintaining mismatch relative to the reference state carries an energetic cost. Whether that sign persists beyond near-equilibrium settings remains open.

VIII. CONCLUSION

We have introduced a variational EFT framework linking quantum information and semiclassical geome-

try through a coarse-grained coherence functional. In controlled limits, its metric variation yields an informational stress tensor that vanishes when the physical state tracks the geometry-adapted reference state and produces leading-order corrections when that mismatch is nonzero.

The paper’s main phenomenological result is an analog-gravity protocol in a BEC sonic horizon. The informational source is fixed by the exact identity $S(\rho||\sigma_\beta) = \beta_H \Delta F$: relative entropy is the free-energy excess over the Hawking-temperature reference. Both coherent and energy-matched thermal injections therefore produce density modulations at the level of $\delta\rho/\rho_0 \sim 10^{-6}$, and the primary observable is their difference, $\Delta A = \kappa_{\text{eff}} \Delta S_{\text{vN}} \approx 2.3 \times 10^{-7}$ at the baseline, which directly measures the von Neumann entropy of the injected occupation and is independent of the Hawking temperature at fixed band occupation. A differential below 5×10^{-8} would falsify the implementation studied in this manuscript. Time-dependent Gross-Pitaevskii simulations validate the density-response kernel at the sub-percent level and quantify the dominant systematic: the state-dependent response of ordinary condensate nonlinearity, which must be discriminated by its spatial and bandwidth signatures.

The framework remains incomplete as a general 3+1D theory. In particular, the coupling scale entering phenomenology is implementation dependent, the sourcing and dissipation sectors await the consistency locking required of hybrid classical–quantum dynamics, and the acoustic calculation is an EFT test of the proposed feedback structure rather than a direct proof of gravitational backreaction in nature. The appropriate claim for the present paper is therefore narrower: it offers a controlled variational ansatz with one falsifiable analog implementation, not a complete new gravity theory.

Appendix A: Variation of the Coherence Functional

A full derivation in 3+1D will require a careful treatment of UV divergences and is left to future work; here we sketch the main steps and assumptions.

In this appendix we derive the informational stress tensor $\Theta_{\mu\nu}$ by varying the coherence functional $\mathcal{C}[g, \rho; \mu]$ with respect to the metric. Recall the definition:

$$\mathcal{C}[g, \rho; \mu] = \int d^4x \sqrt{-g} (\alpha s_{\text{rel}} - \beta s_{\text{tot}} + \gamma \mathcal{R}). \quad (\text{A1})$$

The variation is defined as

$$\delta_g \mathcal{C} = -\frac{1}{2} \int d^4x \sqrt{-g} \Theta_{\mu\nu} \delta g^{\mu\nu}. \quad (\text{A2})$$

¹ Using the identity $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu}$, we have

¹ Strictly speaking, varying the metric also deforms the causal diamond boundary $\partial\mathcal{D}$. We estimate these boundary terms as

generally for any scalar density $\mathcal{L}$:

$$\frac{\delta}{\delta g^{\mu\nu}} \int \sqrt{-g} \mathcal{L} = \sqrt{-g} \left(\frac{\delta \mathcal{L}}{\delta g^{\mu\nu}} - \frac{1}{2} g_{\mu\nu} \mathcal{L} \right). \quad (\text{A3})$$

1. Geometric Contribution

The term $\gamma \mathcal{R}$ with $\mathcal{R} = M_g^2(\mu) R + O(R^2)$ gives the standard Einstein tensor contribution (plus higher order corrections if $\lambda_i \neq 0$):

$$\Theta_{\mu\nu}^{\text{geom}} = \gamma M_g^2(\mu) (G_{\mu\nu} + \dots). \quad (\text{A4})$$

In our convention, the combination $\gamma M_g^2(\mu)$ renormalizes the inverse gravitational coupling on the EFT side of the field equation, while the higher-curvature terms remain subleading counterterms.

2. Relative Entropy Contribution

The relative entropy density is $s_{\text{rel}} = V_{\mathcal{D}}^{-1} \text{tr}(\rho_{\mathcal{D}}(\log \rho_{\mathcal{D}} - \log \sigma_{\mathcal{D}}[g]))$. The metric enters through: 1. The volume factor $V_{\mathcal{D}}$. 2. The definition of the diamond $\mathcal{D}$ (boundary dependence). 3. The reference state $\sigma[g]$.

The variation with respect to the reference state is the most significant new term. For an invertible reference state, the Fréchet derivative of the logarithm gives

$$\delta_{\sigma} S(\rho || \sigma) = -\text{tr}(\rho \delta \log \sigma), \quad (\text{A5})$$

$$\delta \log \sigma = \int_0^{\infty} d\lambda (\sigma + \lambda)^{-1} (\delta \sigma) (\sigma + \lambda)^{-1}. \quad (\text{A6})$$

In near-equilibrium settings where the modular Hamiltonian varies smoothly with the metric, this yields a “reference force” term:

$$\left. \frac{\delta s_{\text{rel}}}{\delta g^{\mu\nu}} \right|_{\sigma} = -\frac{1}{V_{\mathcal{D}}} \text{tr} \left(\rho_{\mathcal{D}} \frac{\delta \log \sigma_{\mathcal{D}}[g]}{\delta g^{\mu\nu}} \right). \quad (\text{A7})$$

Let $K_{\sigma} = -\log \sigma$ be the modular Hamiltonian of the reference state. The leading contribution to the relative entropy variation is then

$$\Theta_{\mu\nu}^{\text{rel}} = \frac{2}{V_{\mathcal{D}}} \left\langle \frac{\delta K_{\sigma}}{\delta g^{\mu\nu}} \right\rangle_{\rho} + (\text{subleading terms}). \quad (\text{A8})$$

so that the overall normalization continues to be carried by the EFT coefficient multiplying $\Theta_{\mu\nu}^{\text{rel}}$ in Eq. (9). If $\sigma[g]$ is locally thermal or vacuum-like, K_{σ} is related to the energy density, and this term represents an informational backreaction. The subleading terms arise from volume variations and boundary effects, which are suppressed by $(L_P/L)^2$.

follows: the area change is $\delta A \sim L^2 \cdot (\delta g) \sim L^2 h$, while the bulk integral scales as L^4 . Thus boundary corrections are $O(L^{-2})$ relative to bulk terms, i.e., suppressed by $(L_P/L)^2$ for $L \gg L_P$. This justifies neglecting them in the macroscopic regime.

3. Generalized Entropy Contribution

The generalized entropy s_{tot} contains the area term $A[\partial \mathcal{D}]/4G$. The variation of the area of the causal diamond boundary with respect to the metric is a standard geometric result. For a diamond defined by proper time τ , the area variation yields a term proportional to the expansion scalar and shear. Combining these, the total informational stress tensor is

$$\Theta_{\mu\nu} = \alpha \Theta_{\mu\nu}^{\text{rel}} - \beta \Theta_{\mu\nu}^{\text{ent}} + \dots \quad (\text{A9})$$

where $\Theta_{\mu\nu}^{\text{ent}}$ includes the area-variation “entanglement pressure”. Explicitly evaluating these requires specifying the map $g \mapsto \sigma[g]$, which we do in the 1+1D model below.

Appendix B: 3+1D Example: Rindler Wedge

To provide a concrete example in 3+1D, consider a Rindler wedge, which approximates the near-horizon geometry of a large causal diamond. The metric in Rindler coordinates $(\tau, \xi, \mathbf{x}_{\perp})$ is $ds^2 = -a^2 \xi^2 d\tau^2 + d\xi^2 + d\mathbf{x}_{\perp}^2$. The geometry-adapted reference state $\sigma[g]$ is the Unruh vacuum $|0_R\rangle$, which appears thermal to an accelerated observer with temperature $T = a/2\pi$.

Consider a probe state ρ that is a single-mode squeezed state of the Rindler scalar field: $\rho = S(\zeta)|0_R\rangle\langle 0_R|S^{\dagger}(\zeta)$, where $\zeta = re^{i\phi}$ is the squeezing parameter. The relative entropy density is non-zero due to the coherence of the squeezed state relative to the thermal background. The informational stress tensor $\Theta_{\mu\nu}$ arises from the variation of this relative entropy with respect to the acceleration parameter a (which encodes the metric). Explicitly, for a massless scalar field, the energy density component is modified by:

$$\langle \Theta_{\tau\tau} \rangle \propto \frac{\delta S(\rho || \sigma)}{\delta g^{\tau\tau}} \sim \sinh^2(r) \cdot T^4. \quad (\text{B1})$$

This demonstrates that even in flat spacetime (viewed by an accelerated observer), coherent excitations (squeezing) induce an effective stress-energy that backreacts on the geometry, consistent with the entropic framework.

Appendix C: 1+1D Polyakov Correspondence

Here we present the details of the 1+1D toy model, including the explicit form of the anomaly-induced stress tensor and its relation to $\Theta_{\mu\nu}$ in Eq. (21).

In two dimensions, the trace anomaly for a conformal field is $\langle T_{\mu}^{\mu} \rangle = \frac{c}{24\pi} R$. This is generated by the Polyakov action:

$$S_P[g] = -\frac{c}{96\pi} \int d^2x \sqrt{-g} \times \int d^2y \sqrt{-g(y)} R(x) G(x, y) R(y), \quad (\text{C1})$$

where $G(x, y)$ is the Green's function for the d'Alembertian $\square$. We identify the geometric part of the coherence functional in 1+1D with this non-local action:

$$\gamma \int \mathcal{R} \rightarrow \gamma S_P[g]. \quad (\text{C2})$$

Varying this with respect to the metric yields the anomaly-induced stress tensor:

$$T_{\mu\nu}^{\text{anom}} = \frac{2}{\sqrt{-g}} \frac{\delta S_P}{\delta g^{\mu\nu}}. \quad (\text{C3})$$

Explicitly, in conformal gauge $ds^2 = -e^{2\omega} du dv$, one finds

$$\langle T_{uu} \rangle_{\text{anom}} = -\frac{c}{12\pi} (\partial_u \omega)^2 + \frac{c}{12\pi} \partial_u^2 \omega. \quad (\text{C4})$$

1. Einstein-Langevin Limit

The entropic equation of motion is

$$G_{\mu\nu} = 8\pi G (\langle T_{\mu\nu} \rangle_\rho + \Theta_{\mu\nu}). \quad (\text{C5})$$

In the limit where ρ is close to $\sigma[g]$, we can expand $\Theta_{\mu\nu}$. If we identify the fluctuations in the relative entropy term with the noise kernel $N_{\mu\nu\alpha\beta}$, we recover the structure of the Einstein-Langevin equation:

$$G_{\mu\nu} = 8\pi G (\langle T_{\mu\nu} \rangle_{\text{ren}} + \xi_{\mu\nu}), \quad (\text{C6})$$

where $\xi_{\mu\nu}$ is a stochastic tensor satisfying $\langle \xi_{\mu\nu} \xi_{\alpha\beta} \rangle \sim N_{\mu\nu\alpha\beta}$. In our framework, $\xi_{\mu\nu}$ arises from the time-dependent mismatch between the actual state $\rho(t)$ and the geometry-adapted reference $\sigma[g(t)]$. This suggests that “quantum noise” in gravity is actually a deterministic signal of informational misalignment.

Appendix D: Explicit Weak-Field Derivation

We derive the explicit form of $\Theta_{\mu\nu}$ for a massless scalar field in a perturbed Minkowski background $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$. The modular Hamiltonian K_σ for the Rindler wedge (approximating a large diamond) is $K_\sigma = 2\pi \int d^3x x T_{00}$. The variation with respect to the metric perturbation $h_{\mu\nu}$ yields:

$$\delta K_\sigma = 2\pi \int d^3x x \delta T_{00}. \quad (\text{D1})$$

For a coherent state ρ , the relative entropy variation is dominated by the expectation value of this modular variation. Thus, the informational stress tensor component Θ_{00} is given by:

$$\Theta_{00} \approx \frac{\alpha}{V_D} \langle \delta T_{00} \rangle_\rho \approx \frac{2\pi\alpha}{V_D} \int d^3k \omega_k \langle a_k^\dagger a_k \rangle_{\text{coh}}. \quad (\text{D2})$$

This explicitly links the informational backreaction to the particle number density of the coherent excitation, scaled by the geometric factor α/V_D .

Numerical estimate: For a coherent state with $\bar{n} = 10^6$ photons at optical frequency ($\omega \sim 1 \text{ eV}$) in a $V_D = 1 \text{ m}^3$ diamond, with $\alpha \sim 1$, the energy density contribution is:

$$\Theta_{00} \sim \frac{\alpha}{V_D} \bar{n} \omega \sim \frac{10^6 \times 1 \text{ eV}}{(1 \text{ m})^3} \sim 7.7 \times 10^{-15} \text{ eV}^4 \sim 7.7 \times 10^{-51} \text{ GeV}^4. \quad (\text{D3})$$

Comparing to the rest mass energy density of a 1 kg test mass in the same volume ($\rho_m \sim 4.3 \times 10^{15} \text{ eV}^4$), this gives:

$$\eta_{\text{coh}} \sim \frac{\Theta_{00}}{\rho_m} \sim 1.8 \times 10^{-30}. \quad (\text{D4})$$

This estimate is meant only as a weak-field dimensional benchmark. It illustrates that, absent the acoustic enhancement discussed below, direct laboratory-scale gravitational effects are tiny.

Appendix E: Analog Gravity Predictions

The paper's main phenomenological target is an acoustic implementation in a Bose-Einstein condensate (BEC) sonic horizon. We do not assume that analog gravity provides Einstein dynamics; rather, we use the acoustic system as a laboratory in which the same state-reference mismatch functional can be coupled to an emergent geometry and interrogated experimentally. The relevant experimental background is the established analog-horizon program initiated by Unruh and developed in BEC systems [24–26].

1. Acoustic Metric in a BEC

In a Bose-Einstein condensate with mass density ρ_0 , sound speed c_s , and flow velocity $\mathbf{v}$, phonon propagation is governed by an emergent acoustic metric [24, 25]:

$$ds_{\text{acoustic}}^2 = \frac{\rho_0}{c_s} [-(c_s^2 - v^2) dt^2 - 2\mathbf{v} \cdot d\mathbf{x} dt + d\mathbf{x}^2]. \quad (\text{E1})$$

A draining vortex configuration creates a sonic horizon where $|\mathbf{v}| = c_s$. This is the acoustic analog of a black hole.

2. Effective Coupling

For the acoustic implementation we package the unresolved microscopic response of the condensate into a *dimensionless* phenomenological coefficient κ_{eff} multiplying the natural hydrodynamic pressure scale $\rho_0 c_s^2$. The relevant microscopic length is the healing length

$$\xi = \frac{\hbar}{\sqrt{2m}c_s} \sim 0.1\text{--}1 \mu\text{m} \quad (\text{E2})$$

for typical ^{87}Rb condensates. Writing the condensate number density as n_0 and the mass density as $\rho_0 = mn_0$, the dimensionless scaling used throughout the numerical model is

$$\kappa_{\text{eff}} \sim \chi_{\text{an}} \frac{1}{n_0 L_{\text{coh}}^3} \left(\frac{\xi}{L_{\text{coh}}} \right)^3, \quad (\text{E3})$$

where χ_{an} is an $O(1)$ calibration constant that packages the unresolved source normalization and L_{coh} is the injected coherence length. Representative values

$$n_0 \sim 10^{14} \text{ cm}^{-3}, \quad \rho_0 \approx 1.4 \times 10^{-5} \text{ kg/m}^3, \quad (\text{E4})$$

$$c_s \sim 1.7 \text{ mm/s}, \quad \xi \sim 0.3 \mu\text{m}, \quad L_{\text{coh}} \sim 10 \mu\text{m} \quad (\text{E5})$$

give

$$\boxed{\kappa_{\text{eff}} \sim 8 \times 10^{-10}} \quad (\text{E6})$$

for $\chi_{\text{an}} \sim 3$. This should be read as the dimensionless scaling choice used in the acoustic ansatz, not as a universal analog of Newton's constant.

3. Informational Stress Tensor for Sonic Horizon

For the acoustic metric Eq. (E1), we evaluate a leading-order informational correction relative to a thermal reference state σ_{acoustic} at the acoustic Hawking temperature set by the horizon surface gravity κ_H ,

$$T_H = \frac{\hbar \kappa_H}{2\pi k_B}, \quad \kappa_H = \left| \frac{\partial}{\partial r} (c_s - v_r) \right|_{r=r_H}. \quad (\text{E7})$$

For a draining bathtub vortex with $v_r = -A/r$ this reduces to $T_H = \hbar A / (2\pi k_B r_H^2)$; steeper engineered horizons reach correspondingly higher T_H . Our baseline takes $\kappa_H = 8.2 \times 10^2 \text{ s}^{-1}$ (gradient length $c_s/\kappa_H \approx 2.1 \mu\text{m}$, comparable to realized steep-horizon configurations [26]), giving $T_H \approx 1.0 \text{ nK}$.

The relative entropy of *any* injected state ρ with respect to the thermal reference σ_β at inverse temperature $\beta_H = 1/k_B T_H$ is fixed exactly by the identity

$$\boxed{S(\rho \parallel \sigma_\beta) = \beta_H (E_\rho - E_\sigma) - (S_{\text{vN}}(\rho) - S_{\text{vN}}(\sigma)) = \beta_H \Delta F}, \quad (\text{E8})$$

i.e., relative entropy is the free-energy excess over the reference in units of $k_B T_H$. Equation (E8) follows directly from $S(\rho \parallel \sigma) = \text{tr}(\rho \log \rho) - \text{tr}(\rho \log \sigma)$ with $\log \sigma = -\beta_H H - \log Z$ and holds for arbitrary ρ .

Two consequences shape the protocol. First, a *coherent* injection is a displaced thermal state: the displacement adds energy $\Delta E = \sum_k |\alpha_k|^2 \hbar \omega_k$ but leaves the entropy unchanged, so

$$\hat{s}_{\text{coh}} \equiv S(\rho_{\text{coh}} \parallel \sigma_\beta) = \beta_H \Delta E = \sum_k |\alpha_k|^2 \frac{\hbar \omega_k}{k_B T_H}. \quad (\text{E9})$$

Second, an *energy-matched thermal* injection that raises the occupation of M modes from $\bar{n}_k$ to $\bar{n}_k + n'$ carries entropy, so

$$\hat{s}_{\text{th}} = \beta_H \Delta E - \Delta S_{\text{vN}}, \quad \Delta S_{\text{vN}} = \sum_k [S_{\text{m}}(\bar{n}_k + n') - S_{\text{m}}(\bar{n}_k)], \quad (\text{E10})$$

with the single-mode entropy $S_{\text{m}}(n) = (1+n) \ln(1+n) - n \ln n$. The thermal control therefore does *not* give a null: at matched injected energy, the coherent and thermal sources differ exactly by the von Neumann entropy of the injected occupation. As a consistency check, for weak thermal injection $n' \ll \bar{n}$ one finds $\hat{s}_{\text{th}} = O(n'^2)$, recovering the standard near-equilibrium quadraticity of relative entropy, because $\ln(1 + 1/\bar{n}_k) = \hbar \omega_k / k_B T_H$ makes the linear terms cancel identically.

The time-time component of the informational stress tensor is modeled as

$$\boxed{\Theta_{tt}^{(\text{analog})}(r) = \rho_0 c_s^2 \kappa_{\text{eff}} \hat{s} g(r)} \quad (\text{E11})$$

where $\hat{s} = S(\rho \parallel \sigma_\beta)$ is the total injected relative entropy in nats, $g(r) = \exp(-|r - r_H|/L_{\text{coh}})$ is the normalized injected-packet envelope, and the normalization to the hydrodynamic pressure scale $\rho_0 c_s^2$ keeps κ_{eff} dimensionless. Compared with an earlier amplitude-squared form of this source, Eq. (E11) implements the relative-entropy functional faithfully: the coherent-versus-thermal asymmetry is now *derived* from Eq. (E8) rather than imposed.

4. Hydrodynamic and Static GP/BdG Benchmarks for the Density Response

The analog system itself is governed by standard condensate dynamics derived from the Gross-Pitaevskii equation and its Bogoliubov linearization [25, 27]. In the long-wavelength hydrodynamic regime $k\xi \ll 1$, the barotropic density response obeys

$$\delta p = c_s^2 \delta \rho, \quad (\text{E12})$$

with $\rho = mn$ the condensate mass density.

We do not claim a full new GP/BdG derivation of the informational source term. Instead, we benchmark the observable relation to the standard compressibility response: if the informational sector contributes a weak quasi-static local source $\Theta_{tt}^{(\text{analog})}$ to the effective Bernoulli balance, then

$$\frac{\delta p}{\rho_0} \simeq \frac{\Theta_{tt}^{(\text{analog})}}{\rho_0}, \quad (\text{E13})$$

and therefore

$$\frac{\delta \rho}{\rho_0} \simeq \frac{\Theta_{tt}^{(\text{analog})}}{\rho_0 c_s^2}. \quad (\text{E14})$$

Equation (E14) is thus benchmarked to standard BEC hydrodynamics.

To benchmark the same response step one level closer to standard condensate theory, consider a locally homogeneous background and let the informational sector enter the stationary Gross-Pitaevskii equation as a weak effective single-particle perturbation

$$\delta U_{\text{info}}(r) \equiv -\frac{\Theta_{tt}^{(\text{analog})}(r)}{n_0}. \quad (\text{E15})$$

Writing $n(r) = n_0 + \delta n(r)$ and linearizing the time-independent GP equation around the background gives

$$\left(g_{\text{BEC}} - \frac{\hbar^2}{4mn_0}\partial_r^2\right)\delta n(r) = -\delta U_{\text{info}}(r), \quad (\text{E16})$$

where $g_{\text{BEC}} = mc_s^2/n_0$ is the standard contact-interaction coefficient expressed in terms of the sound speed. Fourier transforming Eq. (E16) and using $\rho = mn$ and $\xi = \hbar/(\sqrt{2}mc_s)$ yields

$$\frac{\delta\rho_k}{\rho_0} = \frac{\Theta_{tt,k}^{(\text{analog})}}{\rho_0 c_s^2} \frac{1}{1 + \frac{1}{2}(k\xi)^2}. \quad (\text{E17})$$

Equation (E17) is the direct static GP/BdG benchmark used in the numerical model. In the hydrodynamic limit $k\xi \ll 1$, it reduces to Eq. (E14). In addition, we validate the kernel dynamically: imaginary-time relaxation of the full 1D Gross-Pitaevskii equation under weak external potentials reproduces Eq. (E17) to better than 0.6% over $k\xi \in [0.06, 1.5]$ (Fig. 4, left panel). The nonstandard ingredient in the acoustic implementation is therefore the source term Eq. (E11) and the scaling of κ_{eff} , not the linear condensate response once that source is specified.

5. What is Benchmarked and What is Assumed

For clarity, the acoustic implementation contains four logically separate ingredients:

1. **Benchmarked to standard BEC dynamics:** the acoustic metric description, the linear compressibility response Eq. (E14), and the static GP/BdG response kernel Eq. (E17), the latter validated against time-dependent 1D Gross-Pitaevskii relaxation to better than 0.6%.
2. **Derived within the implementation:** the coherent-versus-thermal structure of the source. Given Eq. (E11), the exact identity Eq. (E8) fixes both injections' sources and their difference $\Delta\hat{s} = \Delta S_{\text{vN}}$ with no further freedom; no identification of the control with the reference state is assumed.
3. **Implementation assumption:** the phenomenological scaling of κ_{eff} in Eq. (E6), the proportionality of the source to the injected relative entropy in Eq. (E11), and the envelope profile $g(r)$.

4. **Implementation assumption:** the background flow is held fixed at leading order, and the injected-band occupation (hence ΔS_{vN}) is independently characterized by standard phonon thermometry.

6. Observable Predictions

For each injected state define

$$A \equiv \max_{|r-r_H| \leq L_{\text{coh}}} \left| \frac{\delta\rho}{\rho_0} \right|, \quad \frac{\delta\rho}{\rho_0} = \frac{\Theta_{tt}^{(\text{analog})}}{\rho_0 c_s^2}, \quad (\text{E18})$$

the maximum fractional density modulation within one coherence length of the horizon. The *primary observable* is the coherent-minus-thermal differential at matched injected energy,

$$\Delta A \equiv A_{\text{coh}} - A_{\text{th}} = \kappa_{\text{eff}} \Delta S_{\text{vN}} g_{\text{peak}}, \quad (\text{E19})$$

with $g_{\text{peak}} \approx 0.97$ for the baseline envelope. Equation (E19) has two useful structural properties. First, ΔA depends on the horizon only through $\bar{n}_k(T_H)$ inside ΔS_{vN} , so it is approximately independent of the Hawking temperature at fixed band occupation: the differential directly measures the injected von Neumann entropy. Second, varying the injection bandwidth M at fixed total energy changes ΔS_{vN} (and hence ΔA) while leaving the common-mode energy sourcing unchanged, which provides an internal consistency knob unavailable to energy-proportional backgrounds. If c_s and ξ are varied independently with all other quantities fixed, the explicit sound-speed dependence cancels between Eqs. (E11) and (E14); the physically relevant c_s dependence re-enters only when the Gross-Pitaevskii healing-length relation $\xi = \hbar/(\sqrt{2}mc_s)$ is imposed.

a. Baseline numbers. For the steep-horizon baseline ($T_H \approx 1.0$ nK) with a broadband soft injection centred on $\bar{x} \equiv \hbar\omega/k_B T_H = 2$ ($\bar{\omega} \approx 262$ rad/s, $\bar{k}\xi \approx 0.045$, well inside the hydrodynamic regime), $N_{\text{phonon}} = 10^3$ phonons distributed over $M = 100$ modes ($n' = 10$ per mode, $\bar{n}(\bar{x}) \approx 0.16$), Eqs. (E9)–(E10) give

$$\hat{s}_{\text{coh}} = 2.0 \times 10^3, \quad \Delta S_{\text{vN}} \approx 291, \quad \hat{s}_{\text{th}} \approx 1.71 \times 10^3, \quad (\text{E20})$$

and therefore

$$A_{\text{coh}} \approx 1.57 \times 10^{-6}, \quad A_{\text{th}} \approx 1.34 \times 10^{-6}, \quad \Delta A \approx 2.3 \times 10^{-7}. \quad (\text{E21})$$

Applying the static GP/BdG filter Eq. (E17) to the same source profiles shifts ΔA by 0.63% at the baseline (and by less than 0.9% across the scans below); the response step is thus directly checked against the standard stationary GP/BdG kernel, while the source normalization κ_{eff} remains phenomenological.

Scope of the null test: for the parameter regime in Fig. 5, a differential $\Delta A < 5 \times 10^{-8}$ at $N_{\text{phonon}} \sim 10^3$ would falsify the acoustic implementation defined by Eqs. (E6),

(E8), (E11), and the benchmarked linear-response relations (E14) and (E17), together with the linear-response assumptions above. It would not, by itself, falsify the broader coherence-functional EFT.

7. Robustness of the Observable

We performed one-at-a-time $\pm 25\%$ scans of the differential ΔA about a Gross-Pitaevskii-compatible baseline

$$\begin{aligned} n_0 &= 10^{20} \text{ m}^{-3}, & c_s &\approx 1.7 \text{ mm/s}, & \xi &= 0.3 \mu\text{m}, \\ L_{\text{coh}} &= 10 \mu\text{m}, & T_H &\approx 1.0 \text{ nK}, & \bar{x} &= 2, \\ N_{\text{phonon}} &= 10^3, & M &= 100. \end{aligned} \quad (\text{E22})$$

At the ansatz level, varying n_0 gives $\Delta A \in [1.8 \times 10^{-7}, 3.1 \times 10^{-7}]$; varying ξ gives $\Delta A \in [9.7 \times 10^{-8}, 4.4 \times 10^{-7}]$; varying L_{coh} gives $\Delta A \in [6.0 \times 10^{-8}, 1.3 \times 10^{-6}]$; varying N_{phonon} from 750 to 1250 gives $\Delta A \in [2.1 \times 10^{-7}, 2.5 \times 10^{-7}]$; varying the mode count M gives $\Delta A \in [1.9 \times 10^{-7}, 2.7 \times 10^{-7}]$; and varying T_H at fixed injection band gives $\Delta A \in [2.2 \times 10^{-7}, 2.4 \times 10^{-7}]$, confirming the near- T_H -independence noted above. If one instead imposes the Gross-Pitaevskii healing-length relation $\xi = \hbar/(\sqrt{2}mc_s)$ and varies c_s by $\pm 25\%$, the resulting range is

$$\Delta A \in [1.2 \times 10^{-7}, 5.4 \times 10^{-7}]. \quad (\text{E23})$$

Across these moderate variations, the differential remains above the 5×10^{-8} falsification threshold for the acoustic implementation studied here, with the smallest margin ($\approx 1.2\times$) at the upper end of the L_{coh} scan.

Across the same scans, the static GP/BdG benchmark changes the differential by less than 0.9% relative to the hydrodynamic estimate, with a 0.63% shift at the baseline point. For the present parameter regime, the observable is therefore limited primarily by the implementation assumption for the source term and by the ordinary-nonlinearity systematic quantified below, not by ambiguity in the linear condensate response.

8. Experimental Protocol

1. Create a quasi-2D ^{87}Rb BEC with $N \sim 10^5$ atoms.
2. Engineer a sonic horizon with surface gravity $\kappa_H \sim 10^3 \text{ s}^{-1}$ ($T_H \sim 1 \text{ nK}$), e.g. a steep waterfall step or a tightly imprinted draining vortex.
3. Inject a broadband soft phonon pulse ($\hbar\bar{\omega} \approx 2k_B T_H$, $M \sim 10^2$ modes) via Bragg scattering, alternating between a coherent pulse and a phase-randomized thermal pulse matched in energy and band; characterize the injected occupation (hence ΔS_{vN}) by standard phonon thermometry.

4. Measure the near-horizon density profile with phase-contrast imaging for both injection types, averaging over many dwell times so that the propagating interference background averages down while the static, horizon-pinned component survives.
5. Evaluate the differential $\Delta A = A_{\text{coh}} - A_{\text{th}}$, verify its ΔS_{vN} scaling by repeating at a second bandwidth M at fixed injected energy, and compare with the null threshold 5×10^{-8} .

9. Numerical Simulation

Figure 5 shows an illustrative numerical implementation of the acoustic ansatz together with a one-at-a-time robustness scan of the differential about the Gross-Pitaevskii-compatible baseline above. The code propagates the free-energy source Eq. (E11) through both the standard linear compressibility response and the static GP/BdG kernel Eq. (E17) in order to visualize the coherent and thermal profiles, their difference, and the parameter sensitivity of ΔA . For the representative parameters above, the model gives $A_{\text{coh}} \approx 1.57 \times 10^{-6}$ and $A_{\text{th}} \approx 1.34 \times 10^{-6}$ in the hydrodynamic limit ($\Delta A \approx 2.3 \times 10^{-7}$), with a 0.63% shift of the differential under the static GP/BdG benchmark.

10. Time-Dependent GPE Study: Kernel Validation and the Ordinary-Nonlinearity Systematic

To take the condensate-physics side of the protocol one step beyond static kernels, we solve the time-dependent 1D Gross-Pitaevskii equation (split-step Fourier method, uniform background, $\hbar = m = 1$, $g n_0 = 1$) in two studies (Fig. 4).

Kernel validation. Imaginary-time relaxation under weak external potentials of wavenumber k reproduces the static linear-response kernel Eq. (E17) to better than 0.6% over $k\xi \in [0.06, 1.5]$. The density-response step used for the observable is therefore confirmed directly against the full GP dynamics, not only against its linearization.

Ordinary-nonlinearity confound. We evolve, *without any informational term*, a coherent multi-mode injection (deterministic phases) and an ensemble of phase-randomized injections with the identical amplitude spectrum (energy-matched thermal surrogate), and difference the time-averaged density profiles. For a first-order injection amplitude $\delta n/n_0 \approx 0.1$, the surviving state-dependent differential is at the 10^{-3} – 10^{-2} level—this is the response of ordinary GP nonlinearity to the *same* state property (phase coherence) that the informational source couples to, and it exceeds the informational differential ΔA by several orders of magnitude at baseline injection amplitude. We therefore identify it as the protocol's dominant systematic. Three handles discriminate against it: (i) it is second order in the injected amplitude, so weaker

injection suppresses it quadratically relative to first-order calibration signals; (ii) it rides on the propagating interference pattern, whereas the informational differential is a static profile pinned to the horizon envelope $g(r)$, so averaging over many dwell times suppresses it; and (iii) it varies only weakly with the injection bandwidth M at fixed energy (a factor 1.3 between $M = 4$ and $M = 8$ in the 1D testbed), whereas $\Delta A \propto \Delta S_{vN}(M)$ by Eq. (E19). A full time-dependent simulation of the vortex geometry with both effects present is the natural next step and is left to future work.

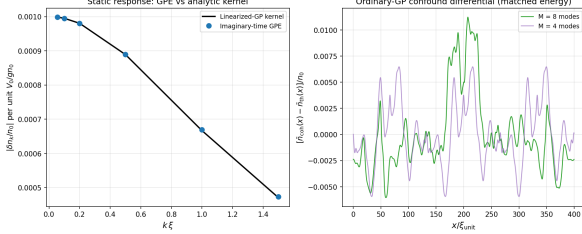


FIG. 4. **Time-dependent 1D GPE study.** Left: static density response measured by imaginary-time GPE relaxation under weak potentials (points) against the analytic linearized kernel of Eq. (E17) (line); maximum deviation 0.53% over $k\xi \in [0.06, 1.5]$. Right: time-averaged density differential between coherent and energy-matched phase-randomized injection evolved under the bare GPE (no informational term) at two bandwidths, quantifying the ordinary state-dependent nonlinear response that constitutes the protocol’s dominant systematic. Reproduced by `gpe_protocol_simulation.py`.

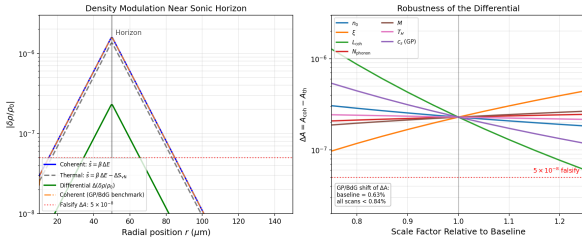


FIG. 5. **Illustrative acoustic-model output.** Left: Fractional density modulation $|\delta\rho/\rho_0|$ versus distance from the horizon for coherent injection ($\hat{s}_{\text{coh}} = \beta_H \Delta E$), the energy-matched thermal injection ($\hat{s}_{\text{th}} = \beta_H \Delta E - \Delta S_{vN}$), their differential, and the static GP/BdG benchmark of the coherent profile. The dotted red line marks the differential falsification threshold 5×10^{-8} . For the plotted baseline, the static GP/BdG response shifts the differential by 0.63% relative to the hydrodynamic estimate. Right: One-at-a-time $\pm 25\%$ robustness scan of the differential $\Delta A = A_{\text{coh}} - A_{\text{th}}$ around the Gross-Pitaevskii-compatible baseline, for n_0 , ξ , L_{coh} , N_{phonon} , the mode count M , T_H at fixed injection band, and c_s with $\xi = \hbar/(\sqrt{2}mc_s)$ enforced for the c_s curve. Within the plotted range, the differential stays above the falsification threshold and the static GP/BdG correction stays below 0.9%. Parameters: ^{87}Rb BEC, $n_0 = 10^{14} \text{ cm}^{-3}$, $\rho_0 \approx 1.4 \times 10^{-5} \text{ kg/m}^3$, $c_s \approx 1.7 \text{ mm/s}$, $\xi = 0.3 \mu\text{m}$, $r_H = 50 \mu\text{m}$, $\kappa_H = 8.2 \times 10^2 \text{ s}^{-1}$ ($T_H \approx 1.0 \text{ nK}$), $\bar{x} = 2$, $L_{\text{coh}} = 10 \mu\text{m}$, $N_{\text{phonon}} = 10^3$, $M = 100$.

Appendix F: Lindblad Generator from Unruh-DeWitt Detector

The Lindblad structure of the decoherence term $\mathcal{L}_g[\rho]$ has been treated as phenomenological in the main text. The present appendix does not derive the full coarse-grained field-theoretic generator used there. Instead, it shows that geometry-dependent GKLS evolution arises in a standard Unruh-DeWitt detector model after weak-coupling and Markovian limits are taken.

1. Setup: Detector Coupled to Field

Consider a two-level system (the “detector”) coupled to a massless scalar field ϕ in curved spacetime:

$$H_{\text{int}} = \lambda \chi(\tau) \mu(\tau) \phi(x(\tau)) \quad (\text{F1})$$

where $\mu(\tau) = |e\rangle\langle g|e^{i\Omega\tau} + \text{h.c.}$ is the detector monopole, $\chi(\tau)$ is a switching function, λ is the coupling, and Ω is the energy gap.

2. Reduced Dynamics

Tracing over the field, the detector density matrix $\rho_D(\tau)$ obeys:

$$\frac{d\rho_D}{d\tau} = -i[H_D, \rho_D] + \mathcal{D}[\rho_D] \quad (\text{F2})$$

where the dissipator, to second order in λ :

$$\mathcal{D}[\rho_D] = \lambda^2 \int_0^\infty ds W(s) \left[\mu(s) \rho_D \mu^\dagger(0) - \frac{1}{2} \{ \mu^\dagger(0) \mu(s), \rho_D \} \right] + \text{h.c.} \quad (\text{F3})$$

and $W(s) = \langle 0 | \phi(x(\tau)) \phi(x(\tau - s)) | 0 \rangle$ is the Wightman function.

3. Geometry Dependence

The Wightman function encodes the spacetime geometry. For a detector at rest in a static spacetime:

$$W(s) = \frac{1}{4\pi^2} \frac{1}{(s - i\epsilon)^2 - |\mathbf{x} - \mathbf{x}'|^2}. \quad (\text{F4})$$

For a uniformly accelerated detector (Rindler):

$$W(s) = -\frac{a^2}{16\pi^2 \sinh^2(a(s - i\epsilon)/2)}. \quad (\text{F5})$$

The thermal character ($T_U = a/2\pi$) emerges from the $\sinh^{-2}$ structure.

4. Lindblad Form

In the Markovian limit ($\Omega\tau \gg 1$), the dissipator takes Lindblad form:

$$\mathcal{D}[\rho_D] = \sum_k \gamma_k \left(L_k \rho_D L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho_D\} \right) \quad (\text{F6})$$

with jump operators and rates:

$$L_- = |g\rangle\langle e|, \quad \gamma_- = \lambda^2 \tilde{W}(\Omega) = \lambda^2 \frac{\Omega}{e^{2\pi\Omega/a} - 1}, \quad (\text{F7})$$

$$L_+ = |e\rangle\langle g|, \quad \gamma_+ = \lambda^2 \tilde{W}(-\Omega) = \lambda^2 \frac{\Omega}{1 - e^{-2\pi\Omega/a}}. \quad (\text{F8})$$

5. Extension to Continuous Systems

For a continuous quantum system (not a two-level detector), we generalize:

$$\mathcal{L}_g[\rho] = \sum_\omega \gamma(\omega, g) \left(A_\omega \rho A_\omega^\dagger - \frac{1}{2} \{A_\omega^\dagger A_\omega, \rho\} \right) \quad (\text{F9})$$

where:

- A_ω are eigenoperators of the free Hamiltonian: $[H_0, A_\omega] = -\omega A_\omega$.
- $\gamma(\omega, g)$ is the transition rate, determined by the Wightman function of the background field.

a. Explicit Form: For a general static metric $ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2$:

$$\gamma(\omega, g) = \lambda^2 \int_{-\infty}^{\infty} ds e^{i\omega s} W_g(s) \quad (\text{F10})$$

which evaluates to:

$$\gamma(\omega, g) = \lambda^2 \frac{|\omega|}{e^{2\pi|\omega|/\kappa_H} - 1} \quad (\text{F11})$$

where $\kappa_H = f'(r_H)/2$ is the surface gravity at the horizon r_H .

6. Connection to the Coherence-Functional EFT

This detector calculation establishes a concrete model system in which:

1. geometry-dependent reduced dynamics takes GKLS/Lindblad form when the field starts in a thermal or vacuum state,
2. The rates $\gamma(\omega, g)$ depend on the metric via the Wightman function.
3. The thermal factor $(e^{\beta\omega} - 1)^{-1}$ with $\beta = 2\pi/\kappa_H$ is generic for horizons.

a. Interpretation. The result motivates the use of a geometry-dependent Lindblad generator in the EFT, but it does not uniquely fix the specific coarse-grained super-operator of Eq. (18). Matching the detector calculation to the modular-Hamiltonian ansatz requires additional coarse-graining assumptions and is left for future work.

Non-Markovian extensions: The Markovian approximation used here is valid for timescales $\tau \gg \Omega^{-1}$, where Ω is the detector gap. For rapidly varying geometries or strong-field regimes, non-Markovian corrections incorporating memory effects are expected to modify the dissipation timescales but not the qualitative coherence-dependence of $\Theta_{\mu\nu}$. A systematic treatment of such corrections will be presented elsewhere.

Appendix G: Local Reference-State Construction and Approximate Uniqueness of $\sigma[g]$

A potential concern is that the map $g \mapsto \sigma[g]$ involves arbitrary choices. In the small-diamond regime used throughout the manuscript, one can identify a natural local Hadamard/KMS reference state. The purpose of this appendix is to show approximate local uniqueness within that regime, not to claim a global uniqueness theorem for arbitrary geometries.

1. Axioms for the Reference State

We require $\sigma_{\mathcal{D}}[g]$ on a causal diamond $\mathcal{D}$ to satisfy:

1. **Hadamard condition:** The two-point function has the standard short-distance singularity structure, ensuring finite renormalized expectation values.
2. **KMS property:** σ is a thermal equilibrium state with respect to the modular flow generated by the diamond's causal structure.
3. **Maximal entropy:** Subject to energy constraints from the local geometry, σ maximizes von Neumann entropy.
4. **Local Lorentz invariance:** In the local inertial frame at the diamond's center, σ reduces to the Minkowski vacuum restricted to the diamond.

2. Approximate Local-Uniqueness Statement (Sketch)

Proposition 1. *For a causal diamond $\mathcal{D}$ of proper size $L \gg L_P$ in a spacetime with curvature radius $R_{\text{curv}} \gg L$, the state $\sigma_{\mathcal{D}}[g]$ satisfying axioms (1)–(4) is unique up to corrections of order $(L/R_{\text{curv}})^2$.*

Proof sketch: By axiom (4), in the local inertial frame the state must approach the Minkowski vacuum. By the Bisognano-Wichmann theorem, this vacuum restricted to

a Rindler wedge (approximating the diamond) is a KMS state at the Unruh temperature $T = a/2\pi$. Axiom (2) then fixes the local reference state to this thermal form up to curvature corrections, while axiom (3) identifies it as the entropy-maximizing state at fixed modular energy. The Hadamard condition (1) is satisfied by construction. Curvature corrections enter at order $(L/R_{\text{curv}})^2$ through the failure of exact boost symmetry. This is therefore an approximate local statement tied to the small-diamond limit used in the main text. $\square$

3. Robustness Under Perturbations

Consider a one-parameter family of reference states $\sigma_\epsilon = \sigma[g] + \epsilon \delta\sigma$ where $\delta\sigma$ is a Hadamard perturbation. Using the Fréchet derivative of the logarithm, the relative entropy varies as:

$$S(\rho||\sigma_\epsilon) = S(\rho||\sigma) - \epsilon \text{tr}(\rho \delta \log \sigma) + O(\epsilon^2). \quad (\text{G1})$$

The leading ambiguity therefore enters through the expectation value of the same linearized operator $\delta \log \sigma$. At equilibrium ($\rho = \sigma$) this ambiguity does not change the matched-state prediction, and near equilibrium it is suppressed by the distance from the reference state. The informational stress tensor $\Theta_{\mu\nu}$ inherits this robustness:

$$\Theta_{\mu\nu}[\sigma_\epsilon] - \Theta_{\mu\nu}[\sigma] = O(\epsilon \|\rho - \sigma\|) + O(\epsilon^2). \quad (\text{G2})$$

This demonstrates that small ambiguities in defining $\sigma[g]$ produce only second-order corrections to physical predictions.

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Competing Interests

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Author Contributions

D.A. is the sole author and was responsible for all aspects of this work including conception, methodology, formal analysis, writing, and visualization.

Data Availability

All data supporting the findings of this study are available within the paper and its supplementary information. The versioned archive containing the manuscript source, simulation code, generated figures, and numerical data is available at <https://doi.org/10.5281/zenodo.17868263> [14]. The development repository is available at <https://github.com/davidahmann/coherism/tree/main/physics>.

Use of AI Tools

Large language models from Anthropic and OpenAI were used for AI-assisted copy editing, literature synthesis, and symbolic manipulation during manuscript preparation. All scientific judgment, physical interpretation, derivations, and responsibility for claims made in this work are the author's own. The author has reviewed and approved all content and confirms it reflects the original work.

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